HW 7: Simple Harmonic Oscillator

This homework walks you through S&N Section 2.3. The Harmonic Oscillator Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}, \qquad [\hat{x}, \hat{p}] = i\hbar. \tag{1}$$

We define the annihilation operator a, the creation operator $a^\dagger$, and the number operator N as

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right), \qquad \hat{N} = \hat{a}^\dagger\hat{a}. \tag{2}$$

The eigenkets of $\hat{N}$ are labeled by their eigenvalue n, $\hat{N}\,|n\rangle = n\,|n\rangle$, and are normalized, $\langle n|n\rangle = 1$.

Recall that $[\hat{a}, \hat{a}^\dagger] = 1$ and $\hat{N} = \hat{a}^\dagger\hat{a}$, such that

$$\hat{H} = \hbar\omega\left(\hat{N} + \frac{1}{2}\right). \tag{3}$$

such that $\hat{H}\,|n\rangle = E_n\,|n\rangle$ with $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$.

a) Show that $[\hat{N}, \hat{a}] = -\hat{a}$ and $[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$.

b) Show that

$$\hat{N}\left(\hat{a}\,|n\rangle\right) = (n-1)\left(\hat{a}\,|n\rangle\right), \qquad \hat{N}\left(\hat{a}^\dagger\,|n\rangle\right) = (n+1)\left(\hat{a}^\dagger\,|n\rangle\right), \tag{4}$$

In other words, $\hat{a}$ lowers the eigenvalue of $\hat{N}$ by one, and $\hat{a}^\dagger$ raises it by one, justifying the names "annihilation" and "creation" come from.

Note that $n \geq 0$ by writing $n = \langle n|\,\hat{N}\,|n\rangle$ as the squared norm of the ket $\hat{a}\,|n\rangle$. The lowest eigenvalue is $n = 0$ with $\hat{a}\,|0\rangle = 0$ and $n = 0, 1, 2, ...$; the spectrum is discrete.

c) Given that $\hat{a}\,|n\rangle = c\,|n-1\rangle$ for some number c. Compute $|c|^2$ by taking the norm of both sides, and choose c real and positive. Do the same for $\hat{a}^\dagger\,|n\rangle$. You should find

$$\hat{a}\,|n\rangle = \sqrt{n}\;|n-1\rangle, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\;|n+1\rangle. \tag{5}$$

d) Inverting the definitions of a and $a^\dagger$, we get

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right), \qquad \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}\left(-\hat{a} + \hat{a}^\dagger\right). \tag{6}$$

What can you conclude about the expectation value of position for any number state $|n\rangle$ in the harmonic oscillator? Is this always true for superpositions of number states?

e) Compute $\langle n|\,\hat{x}^2\,|n\rangle$ and $\langle n|\,\hat{p}^2\,|n\rangle$. *Hint:* in $(a+a^\dagger)^2$, only the terms with one a and one $a^\dagger$ survive, and $aa^\dagger + a^\dagger a = 2N + 1$. Show that $\Delta x\,\Delta p = \left(n + \frac{1}{2}\right)\hbar$ in the state $|n\rangle$. Which state saturates the uncertainty principle?

f) Now we find the ground-state wave function $\langle x|0\rangle$. Project $a\,|0\rangle = 0$ onto $\langle x|$ and use $\langle x|\,p\,|\alpha\rangle = -i\hbar\frac{d}{dx}\,\langle x|\alpha\rangle$ to obtain the first-order differential equation

$$\left(x + x_0^2\frac{d}{dx}\right)\langle x|0\rangle = 0, \tag{7}$$

where $x_0 = \sqrt{\frac{\hbar}{m\omega}}$. Find the normalized solution

$$\psi_0(x) = \langle x|0\rangle = \frac{1}{\pi^{1/4}\sqrt{x_0}} \exp\left[-\frac{1}{2}\left(\frac{x}{x_0}\right)^2\right]. \tag{8}$$

Note then that using $|1\rangle = a^\dagger |0\rangle$, we get

$$\langle x|1\rangle = \sqrt{2}\,\frac{x}{x_0}\,\langle x|0\rangle\,. \tag{9}$$

and similar for higher excited states, e.g.,

$$\langle x|2\rangle = \frac{1}{\sqrt{2}}\left(2\frac{x^2}{x_0^2} - 1\right)\langle x|0\rangle\,. \tag{10}$$

g) Finally, the time evolution. Use the Heisenberg equation of motion to show that, in the Heisenberg picture,

$$\frac{da}{dt} = -i\omega a \qquad \Longrightarrow \qquad a(t) = a(0)\,e^{-i\omega t}. \tag{11}$$

and that

$$\hat{x}(t) = \hat{x}(0)\cos\omega t + \frac{\hat{p}(0)}{m\omega}\sin\omega t, \tag{12}$$

which is the classical solution, with operators in place of numbers. Why is $\langle\hat{x}\rangle$ zero at all times even though the operator $\hat{x}(t)$ oscillates in time?

Bonus) Let $\hat{x}$ and $\hat{p}$ denote Schrödinger-picture operators. Show that

$$\hat{A}(t) = \hat{x}\cos\omega t - \frac{\hat{p}}{m\omega}\sin\omega t$$

is a conserved observable. For a fixed real parameter λ, define

$$\hat{W}_\lambda(t) = e^{-i\lambda\hat{A}(t)/\hbar}.$$

Determine $\hat{W}_\lambda^\dagger\,\hat{x}\,\hat{W}_\lambda$ and $\hat{W}_\lambda^\dagger\,\hat{p}\,\hat{W}_\lambda$ to order $\mathcal{O}(\lambda)$.

Solutions

For a, b, c and f, please see S&N Section 2.3. The time evolution in (g) is also discussed there.

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d) Acting a and $a^\dagger$ on number states changes n:

$$\langle n|\,\hat{x}\,|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n}\,\langle n|n-1\rangle + \sqrt{n+1}\,\langle n|n+1\rangle\right) = 0, \tag{13}$$

and similarly for $\langle n|\,\hat{p}\,|n\rangle = 0$. This is not true for superpositions. For example, for $|\psi\rangle = c_0\,|0\rangle + c_1\,|1\rangle$, the cross terms survive:

$$\langle\psi|\,\hat{x}\,|\psi\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(c_0^* c_1\,\langle 0|\,\hat{a}\,|1\rangle + c_1^* c_0\,\langle 1|\,\hat{a}^\dagger\,|0\rangle\right) = \sqrt{\frac{2\hbar}{m\omega}}\,\mathrm{Re}\left(c_0^* c_1\right), \tag{14}$$

which is non-zero unless $c_0^* c_1$ is purely imaginary. For example, the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ has

$$\langle\psi|\,\hat{x}\,|\psi\rangle = \sqrt{\frac{\hbar}{2m\omega}},$$

but $|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ has

$$\langle\phi|\,\hat{x}\,|\phi\rangle = 0,$$

Note how this example shows that the relative phase between the states in a superposition is indeed physical. You may want to try drawing the two wavefunctions (recall $x_0 = \sqrt{\hbar/m\omega}$):

$$|\psi(x)|^2 = \frac{e^{-x^2/x_0^2}}{\sqrt{\pi}\,x_0}\left(\frac{1}{2} + \sqrt{2}\,\frac{x}{x_0} + \frac{x^2}{x_0^2}\right), \quad |\phi(x)|^2 = \frac{e^{-x^2/x_0^2}}{\sqrt{\pi}\,x_0}\left(\frac{1}{2} + \frac{x^2}{x_0^2}\right). \tag{15}$$

Note how only ψ ends up with a odd term linear in x, shifting the expectation value of $\hat{x}$ away from zero.



By the way, what happens with $\langle\hat{p}\rangle_\psi$ and $\langle\hat{p}\rangle_\phi$?

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e) Using the hint,

$$\langle n|\,\hat{x}^2\,|n\rangle = \frac{\hbar}{2m\omega}\,\langle n|\,(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a})\,|n\rangle = \frac{\hbar}{2m\omega}(2n+1), \tag{16}$$

$$\langle n|\,\hat{p}^2\,|n\rangle = -\frac{m\hbar\omega}{2}\,\langle n|\,(-\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a})\,|n\rangle = \frac{m\hbar\omega}{2}(2n+1), \tag{17}$$

where the terms $\hat{a}^2$ and $\hat{a}^{\dagger 2}$ drop out because they change n by two units. Since $\langle\hat{x}\rangle_n = \langle\hat{p}\rangle_n = 0$, the variances are just $\Delta\hat{x}^2 = \langle\hat{x}^2\rangle_n$ and $\Delta\hat{p}^2 = \langle\hat{p}^2\rangle_n$ in the number state $|n\rangle$. So

$$\Delta x\,\Delta p = \sqrt{\langle\hat{x}^2\rangle_n\,\langle\hat{p}^2\rangle_n} = \left(n + \frac{1}{2}\right)\hbar. \tag{18}$$

Only the ground state, $n = 0$, saturates $\Delta x\,\Delta p \geq \hbar/2$.



As a bonus, notice that the kinetic and potential energies are equal on average, that is $\langle\hat{p}^2\rangle/2m = \frac{1}{2}m\omega^2\langle\hat{x}^2\rangle = \frac{1}{2}E_n$. This is just the virial theorem for our quadratic potential.

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g) In the Heisenberg picture, $\frac{d\hat{a}}{dt} = \frac{1}{i\hbar}\left[\hat{a}, \hat{H}\right]$. Using $\hat{H} = \hbar\omega(\hat{N} + \frac{1}{2})$ and $[\hat{a}, \hat{N}] = \hat{a}$,

$$\frac{d\hat{a}}{dt} = \frac{\hbar\omega}{i\hbar}\left[\hat{a}, \hat{N}\right] = -i\omega\,\hat{a} \qquad \Longrightarrow \qquad \hat{a}(t) = \hat{a}(0)\,e^{-i\omega t} \tag{19}$$

Plugging this and its hermitian conjugate into the expression for $\hat{x}$ in part (d),

$$\begin{aligned}\hat{x}(t) &= \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}(0)\,e^{-i\omega t} + \hat{a}^\dagger(0)\,e^{i\omega t}\right) \\ &= \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}(0) + \hat{a}^\dagger(0)\right)\cos\omega t + \sqrt{\frac{\hbar}{2m\omega}}\,i\left(\hat{a}^\dagger(0) - \hat{a}(0)\right)\sin\omega t = \hat{x}(0)\cos\omega t + \frac{\hat{p}(0)}{m\omega}\sin\omega t,\end{aligned} \tag{20}$$

where we used $\hat{p}/m\omega = i\sqrt{\hbar/2m\omega}\,(\hat{a}^\dagger - \hat{a})$. Similarly, $\hat{p}(t) = \hat{p}(0)\cos\omega t - m\omega\,\hat{x}(0)\sin\omega t$. These are just the classical solutions for the harmonic oscillator written in terms of the operators (in the Heisenberg picture).



Note that for a number state we still have

$$\langle n|\,\hat{x}(t)\,|n\rangle = \langle n|\,\hat{x}(0)\,|n\rangle\cos\omega t + \frac{\langle n|\,\hat{p}(0)\,|n\rangle}{m\omega}\sin\omega t = 0. \tag{21}$$

The operator $\hat{x}(t)$ does oscillate, but the average value of $\hat{x}(t)$ vanishes for a number state since it vanished at $t = 0$, and the number state is a stationary state! Only expectation values of superpositions of neighboring number states exhibit oscillatory behavior $\langle n \pm 1|\,\hat{x}(t)\,|n\rangle \propto e^{\pm i\omega t}$. This is not in contradiction with the fact that $\hat{x}(t)$ oscillates; it simply reflects the fact that the expectation value of a stationary state remains constant.

Bonus) First note that I say x and p are Schrödinger-picture operators, i.e., they are constant in time and we might as well call them $\hat{x} = \hat{x}(0)$ and $\hat{p} = \hat{p}(0)$. The new operator $\hat{A}(t)$ is a Schrödinger-picture operator with an explicit time dependence, so it is conserved if its Heisenberg-picture version, $\hat{A}_H(t) = \hat{U}^\dagger(t)\hat{A}(t)\hat{U}(t)$, is time independent, or equivalently, if

$$\frac{d\hat{A}_H}{dt} = \hat{U}^\dagger\left(\frac{\partial\hat{A}}{\partial t} + \frac{1}{i\hbar}\left[\hat{A}, \hat{H}\right]\right)\hat{U} = 0. \tag{22}$$

Using $[\hat{x}, \hat{H}] = i\hbar\hat{p}/m$ and $[\hat{p}, \hat{H}] = -i\hbar m\omega^2\hat{x}$,

$$\frac{\partial\hat{A}}{\partial t} + \frac{1}{i\hbar}\left[\hat{A}, \hat{H}\right] = \left(-\omega\hat{x}\sin\omega t - \frac{\hat{p}}{m}\cos\omega t\right) + \left(\frac{\hat{p}}{m}\cos\omega t + \omega\hat{x}\sin\omega t\right) = 0. \tag{23}$$

The explicit time dependence exactly cancels the change due to the dynamics. A quicker way to see this is to use Eq. (20) directly:

$$\hat{A}_H(t) = \hat{x}(t)\cos\omega t - \frac{\hat{p}(t)}{m\omega}\sin\omega t = \hat{x}\left(\cos^2\omega t + \sin^2\omega t\right) + \frac{\hat{p}}{m\omega}\left(\sin\omega t\cos\omega t - \sin\omega t\cos\omega t\right) = \hat{x}. \tag{24}$$

In other words, $\hat{A}(t)$ is the operator that "evolves the position back to its initial configuration". In this sense, the conserved quantity here is the initial position itself (not a very deep insight, but a true statement nonetheless).

For the transformation, we use $e^{\hat{B}}\hat{O}e^{-\hat{B}} = \hat{O} + [\hat{B}, \hat{O}] + \frac{1}{2!}[\hat{B}, [\hat{B}, \hat{O}]] + \cdots$ with $\hat{B} = i\lambda\hat{A}/\hbar$, since $\hat{W}_\lambda^\dagger = e^{i\lambda\hat{A}/\hbar}$. The commutators are all proportional to the identity

$$\left[\hat{A}, \hat{x}\right] = -\frac{\sin\omega t}{m\omega}\,[\hat{p}, \hat{x}] = \frac{i\hbar}{m\omega}\sin\omega t, \qquad \left[\hat{A}, \hat{p}\right] = \cos\omega t\,[\hat{x}, \hat{p}] = i\hbar\cos\omega t, \tag{25}$$

so all the nested commutators vanish and the $\mathcal{O}(\lambda)$ result is in fact exact:

$$\hat{W}_\lambda^\dagger\, \hat{x}\, \hat{W}_\lambda = \hat{x} + \frac{i\lambda}{\hbar}\left[\hat{A}, \hat{x}\right] = \hat{x} - \frac{\lambda}{m\omega}\sin\omega t, \qquad \hat{W}_\lambda^\dagger\, \hat{p}\, \hat{W}_\lambda = \hat{p} + \frac{i\lambda}{\hbar}\left[\hat{A}, \hat{p}\right] = \hat{p} - \lambda\cos\omega t. \quad (26)$$

The transformation shifts position and momentum by $\delta x(t) = -\frac{\lambda}{m\omega}\sin\omega t$ and $\delta p(t) = -\lambda\cos\omega t$. At $t = 0$, $\hat{W}_\lambda(0) = e^{-i\lambda\hat{x}/\hbar}$ is simply a momentum kick by $-\lambda$ which leads to a shift in the time evolution of the system described by Eq. (26).



The operator $\hat{W}_\lambda$ is called the displacement operator (hopefully the name is self-explanatory). Note that $\hat{W}_\lambda$ is a symmetry of the oscillator, even though it does not commute with $\hat{H}$ (similar to what we saw with Galilean boosts). There is a lot more to this story that shows how this transformation produces **coherent states**. Coherent states behave as close to classical point particles in phase space as possible in quantum mechanics. They are defined as eigenstates of the annihilation operator $\hat{a}$, and can be generated by acting with the displacement operator $\hat{W}_\lambda$ on the ground state of the harmonic oscillator:

$$|\alpha\rangle = \hat{W}_\alpha\, |0\rangle\,, \quad (27)$$

where α is a complex number parameterizing the displacement in phase space. You can check that $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$ and that these states have $\Delta x \Delta p = \frac{\hbar}{2}$, which is the closest one can get to a classical system in QM. Coherent states are extremely important in quantum optics and information theory.